

Mohammed Afrah Usman

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SUMMARY

Computer Science Engineering student with hands-on experience building production grade AI systems, full-stack applications, and real-time embedded pipelines. Delivered a Learning-to-Rank search engine, a privacy-preserving LLM code consultation framework, and an IoT sports platform backed by a filed Indian patent and a Government of Karnataka grant. Skilled in Python, FastAPI, React, and cloud technologies across end-to-end system development.

EDUCATION

Mangalore Institute of Technology & Engineering
Bachelor of Engineering in Computer Science

Sep 2023 – Jun 2027
Karnataka, India

- CGPA: 8.6/10.0 with coursework and projects focused on software engineering, AI, and cloud technologies.

SKILLS

Programming Languages: Java, Python, C, JavaScript, HTML, CSS, TypeScript

Frameworks & Libraries: React, FastAPI, Flutter, Tailwind CSS

Databases: MySQL, Firebase, PostgreSQL, Supabase

AI & Machine Learning: LightGBM, BM25, Scikit-learn

Developer Tools: Git, GitHub, Docker, Vercel, Postman, Pytest, Jira

Cloud Technologies: Google Cloud Platform (GCP), Amazon Web Services (AWS)

Other: REST APIs, Embedded Systems (Arduino, ESP32, Raspberry Pi), CI/CD, Linux

EXPERIENCE

Vyanika Innovations LLP (Tech Kabaddi) | Startup
Software & Product Development Engineer

May 2025 – Apr 2026

- Built a real-time court visualisation interface on Raspberry Pi 5 using ESP32 and UDP socket communication, achieving sub-200ms end-to-end display latency.
- Developed Python event processing modules across an Arduino Mega + ESP32 + Raspberry Pi 5 embedded pipeline, validated at 96–98% detection accuracy in live player simulation testing.
- Co-developed an IoT sports technology platform selected for the Bolpu Acceleration Program and backed by a filed Indian patent under Vyanika Innovations LLP.

PROJECTS

CodeAbstract – Privacy Framework for Secure LLM Code Consultation | Group of 4

Mar 2026 – Present

- **Tech Stack:** Python, AST, Typer, Cryptography, Pytest, Git
- Built a privacy-preserving framework for secure LLM-assisted code review using AST-based semantic abstraction and bidirectional mapping.
- Developed an end-to-end pipeline for code abstraction, automated LLM interaction, reverse translation, and context-aware feedback reconstruction.
- Designed a modular system to protect source code, API keys, and business logic while preserving semantic structure for AI analysis.

LTR Search Engine

Mar 2026 – May 2026

- **Tech Stack:** Python, LightGBM, BM25, FastAPI, Streamlit
- Built a Learning-to-Rank search engine using BM25 retrieval and LambdaMART reranking, achieving NDCG@10 of 0.8665 on the MSLR-WEB10K dataset.

- Trained a LightGBM model on a 136-feature ranking pipeline and deployed a real-time search application with 58ms end-to-end latency.

AI Data Translation Layer (AITL)

Feb 2026 – April 2026

- **Tech Stack:** Python, FastAPI, Gemini AI, PostgreSQL, React, Docker
- Built a data intelligence engine converting unstructured PDF, CSV, and TXT files into structured AI-ready datasets via a 7-stage pipeline.
- Achieved 80–85% automated data cleaning accuracy, reducing manual preprocessing effort by 80%.

Imchef E-Commerce Platform

Sep 2025 – Oct 2025

- **Tech Stack:** React, Firebase, Google Cloud, Vercel, EmailJS
- Developed and deployed a full-stack e-commerce platform with Firebase authentication, dynamic product listings, and automated order workflows via EmailJS.

CERTIFICATIONS

Feature Engineering – Kaggle	May 2026
Introduction to Soft Computing – NPTEL	Mar 2026
Postman API Fundamentals Student Expert – Postman	Jul 2024
Figma Essential for UI/UX – Udemy	Dec 2024

ACHIEVEMENTS

- Filed Indian patent contributor Application No. 202641005974 (Indian Patent Office, 2026) for an IoT-based smart sports officiating system under Vyanika Innovations LLP.
- K-Tech NAIN 2.0 grant recipient (Government of Karnataka) Rs. 5,00,000 sanctioned for hardware deeptech innovation; 99.8% utilisation across a structured 9-phase development cycle.
- DPIIT-recognised startup contributor (Certificate No. DIPP245855, Feb 2026) – Vyanika Innovations LLP recognised under Startup India in the Technology Hardware and Electronics sector.
- Selected for Bolpu Acceleration Program (Karnataka startup ecosystem) for Tech Kabaddi, an IoT-based sports technology system registered under Vyanika Innovations LLP.
- Won 1st Place in the Web Development Competition at AAKAR 2026.
- Earned Champions Tier in the Google Cloud Arcade Facilitator Program with 78 Arcade Points.
- Qualified for Round 2 of TCS TechBytes 2026, a statewide engineering IT quiz by Tata Consultancy Services.